Fullstack Project: Home Building Regulatory Engine

Overview

Project Statement

One of the key early questions for any residential construction project is simple but difficult to answer:

> ℹ️ For a given a parcel, what can I confidently build?

While zoning codes and regulatory documents exist, they are distributed across multiple sources and require interpretation to translate general rules into parcel-specific constraints. As projects evolve and new project inputs emerge, this understanding must be continuously refined rather than treated as a one-time determination.

The core challenge is producing an evidence-backed buildability assessment that synthesizes these sources into a clear statement of what is allowed, the confidence in that conclusion, and the specific data and rules that support it. When regulations are ambiguous or incomplete, the system should also surface reasonable interpretations or decision paths while clearly distinguishing them from verified constraints.

Business Context & Impact

At Cover, every project begins with determining what can realistically be built on a site, yet this process is often slow and manual due to fragmented regulatory research, creating uncertainty that affects early design exploration, client conversations, pricing, and timelines. A reliable, evidence-backed buildability assessment would allow us to provide faster and more transparent feasibility

guidance, helping clients make informed decisions earlier while reducing entitlement surprises and building trust in the process.

Ultimately, transforming regulatory complexity into structured, explainable information supports Cover's mission to make high-quality homebuilding more predictable, accessible and the best overall client experience. In the short term, this will massively speed up the design team's ability to assess property buildability throughout the project.

Key Impact Metrics

- Reduce regulatory research effort to zero
- Answer buildability questions in real-time as inputs change
- Consistent cited regulatory answers across parcels

Technical Requirements

Required Programming Languages: Python backend, Javascript/Typescript frontend

AI/ML Frameworks: Up to the candidate

Development Tools: Up to the candidate

Cloud Platforms: AWS preferred, GCP may be needed for geocoding

Other Specific Requirements: Up to the candidate (recommend: postgres, postgis, shapely)

Target User

The initial target user is an Architect or Engineer with knowledge of the code and regulations. They should be able to validate the claims made by the tool. They know how to read a map, understand parcels, and understand technical regulatory language.

Scope

More specific resources will be provided before the project starts to alleviate some of the guessing on what sources are best.

- Proof of Concept
- Use the latest in LLM/AI/ML technology to support the underlying solution
- Geographic scope limited to residential parcels within the City of Los Angeles as a representative test environment
- Use publicly available parcel, zoning, and regulatory information where feasible (or reasonable mock data if needed)
- Emphasize regulatory ingestion and synthesis as the core focus, demonstrating how municipal zoning text can be transformed into structured, explainable rule fragments that support downstream parcel reasoning.
    - Implement a small, well-reasoned subset of rules and/or agencies rather than full regulatory coverage
- Demonstrate parcel to constraints reasoning with explicit citations and confidence signals
- Emphasize a flexible architecture designed to scale across LA City, LA County, other California jurisdictions, and eventually broader regions beyond California
- Prioritize clarity, traceability, and explainability over completeness of the regulatory modeling

Deliverables

Time Constraints: 1 week

Code Quality Expectations: Use your judgement

Performance Benchmarks: Use your judgement

Functional Requirements (Must-Haves)

- An interface to enter a residential address or APN, and get a real-time structured buildability assessment for the parcel.
    - It should identify applicable zoning and regulatory constraints, include supporting evidence and citations, and clearly distinguish deterministic

parcel data from interpretive or inferred conclusions.

    - The output should include a confidence signal and a transparent explanation of how each constraint was derived.
- Some basic visualization of the parcel and any existing buildings if any.
- Tested on a range of *provided* residential parcels/addresses and various building types (Single Family Home, ADU, Guest House, etc.)
- Ability for a target user to interact with the interface and easily see the outputs
- Simple but actionable Architecture diagram illustrating how this could be productionized

Bonus

- The output should be responsive to other project related inputs and consider the current state of the project. This can be simplified to descriptive or parameterized inputs. (i.e. 1 bedroom, 1 bathroom, etc).
- Accept some form of feedback from target users on bad responses to improve future responses
- Chat-like interface to ask follow-up questions about the results and underlying dataset
- Interactive map translating regulations into visual annotations for better understanding
- Generate actionable geometry like property “setbacks” based on the parcel information and synthesized regulations
- Admin interface to demonstrate the regulatory engine pipeline, adjust system, etc.